Contribution Report: YOLOv11 Multiclass Animal Image Classifier

MOHAN DEV ROY

ID-2223088042

Project overview

The project classifies animal images into five classes (Dog, Cow, Cat, Lamb, Zebra) using two models: a Sequential CNN and YOLOv11. My part was training and evaluating the YOLOv11 model; the YOLO dataset was created and preprocessed by my teammate Sudarshan Ghosh Durjoy.

Team contributions

The work on the YOLO model was split in two: MY Teammate Sudarshan Ghosh Durjoy built the dataset, and I trained and evaluated the model on it.

Contributor	Contribution
Durjoy	Created the YOLO dataset and carried out its preprocessing
Me	Trained the YOLOv11 classifier on Durjoy's dataset, evaluated it, and recorded the results (94% accuracy)

Open question: list the exact preprocessing steps Durjoy applied (resizing, labelling, augmentation, cleaning)

Dataset

The dataset states 500 images across five classes, 100 per class.

Images per class:

Class	Total
Dog	100
Cow	100
Cat	100
Lamb	100
Zebra	100

YOLOv11 training setup

I trained a YOLOv11 image classifier on Durjoy's dataset.

Setting	Value
Model	YOLOv11 classification model,
Task	Five-class image classification
image size	640X640
- Training set (70%): 350 images - Validation set (20%): 100 images - Test set (10%): 50 images	

Results

The YOLOv11 classifier reached 94% accuracy.

Model	Accuracy
YOLOv11 (my contribution)	94%

Conclusion and limitations

Durjoy's YOLO dataset let me train a YOLOv11 classifier that reaches 94% accuracy on the five animal classes, meeting the project's 90% target.